

ФОРМУЛА НЬЮТОНА ДЛЯ ИНТЕРПОЛЯЦИИ

$$p(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + \dots + a_n(x-x_0)\dots(x-x_{n-1})$$

a_k - РАЗДЕЛЁННЫЕ РАЗНОСТИ

$$f[x_i] = y_i;$$

$$f[x_i, x_j] = \frac{y_j - y_i}{x_j - x_i}$$

$$f[x_i, x_j, x_k] = \frac{f(x_j, x_k) - f(x_i, x_j)}{x_k - x_i}$$

$$a_0 = f(x_0)$$

$$a_1 = f(x_0, x_1)$$

$$a_2 = f(x_0, x_1, x_2)$$

...

ПРИМЕР: (1; 2), (2; 3), (4; 1)

X	f(x)	f(x ₁ , x ₂)	f(x ₁ , x ₂ , x ₃)
1	2	$f(1, 2) = \frac{3-2}{2-1} = 1$	$f(1, 2, 4) = \frac{f(2, 4) - f(1, 2)}{4-1} = \frac{-1-1}{3} = -\frac{2}{3}$
2	3		
4	1	$f(2, 3) = \frac{1-3}{4-2} = -1$	